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Integrating Local AI Agents within the openSUSE Ecosystem

From cloud chatbots to self-improving agents that live on your Linux system

Maulana Abdurrahman

Yogyakarta | October, 2026

About



- Maulana Abdurrahman
- openSUSE user
- Senior GIS Developer
- Jakarta

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01 Paradigm shift & the privacy problem

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The Paradigm is Shifting

BEFORE

Stateless cloud chatbots

- Session resets every time
- Heavy reliance on proprietary APIs
- Data leaves your machine
- Limited offline / system automation
- No lasting procedural memory

NOW

Autonomous local-first agents

- Persistent across sessions
- Open-source orchestration layer
- Skills & memory stay on disk
- Can drive real OS workflows
- Self-improving via Learning Loop

Why Local-First Matters

Data Privacy

Prompts, code, configs, and system state often leave your machine when using pure cloud APIs.

Offline & Reliability

Cloud-only agents fail when the network or vendor is unavailable.

System Automation

True OS-level workflows need an agent that can live next to your files, services, and scripts.

Vendor Lock-in

Proprietary orchestration layers make it hard to switch models or keep full control.

02 What is Hermes Agent?

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What is Hermes Agent?

Open-source autonomous agent by Nous Research (MIT license)

Closed Learning Loop

Creates & improves skills from experience

Persistent Memory

Remembers across sessions & restarts

Model-Agnostic

OpenRouter, OpenAI, Anthropic, local, any endpoint

Runs on Your Infra

CLI, Desktop, 20+ messaging platforms

Skills as Markdown

Procedural knowledge as plain files on disk

Tool-rich

Terminal, files, browser, code, cron, MCP...

03 Core architecture on openSUSE

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Core Architecture on openSUSE

This talk uses cloud LLM backends (OpenRouter / OpenAI / Anthropic / etc.) — not local Ollama.
Environment runs directly on the host (no Podman/sandbox isolation required for the demo).

Interface Layer

CLI • Desktop App • Telegram / Discord / Slack / ...

Hermes Agent Core

Planner • Tool Executor • Memory • Skill Manager • Learning Loop

LLM Backend (Cloud / Local)

OpenRouter / OpenAI / Anthropic / Nous Portal / Ollama / any compatible endpoint

openSUSE Host

Filesystem (~/.hermes/skills) • Terminal • Services • User workflows

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04 Learning Loop deep dive + Markdown skills

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The Closed Learning Loop

Hermes is the only major agent with a built-in learning loop that turns experience into reusable skills.

1

Observe

Agent performs multi-step tasks; every tool call & correction is tracked

2

Distill

Background review decides a workflow is worth repeating → writes SKILL.md

3

Reuse

Next time the skill is loaded (progressive disclosure) → faster, cheaper

4

Refine

Skills self-improve during use; you can also edit them by hand

Learning Loop — How Skills Are Written

Trigger	After a complex task, after errors that were fixed, or after you correct the agent. No numeric confidence threshold — the model decides.
Mechanism	Background review (auxiliary.background_review) forks after turns and may call skill_manage (create / patch / edit).
Output	A directory under ~/.hermes/skills/ containing a SKILL.md (and optional scripts/, references/, templates/).
Progressive disclosure	Model first sees only metadata of all skills (~3k tokens). Full body loaded only when the skill is selected.
You stay in control	skills.write_approval can require human approval before a skill is written. Skills are plain Markdown — edit with any text editor.

SKILL.md - Anatomy of a Learned Skill

Directory layout

```
~/.hermes/skills/
  category/
    skill-name/
      SKILL.md          <- required
      scripts/          <- optional
      references/       <- optional
      templates/        <- optional
      assets/           <- optional
```

SKILL.md structure

```
---
name: weekly-backup
description: ...
version: 1.0.0
metadata:
  hermes:
    tags: [sysadmin]
---
# Title
## When to Use
## Procedure
## Pitfalls
## Verification
```

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Managing Skills on openSUSE Filesystem

Edit with any editor

vim, nano, kate, VS Code — skills are ordinary Markdown files.

Version control

Put ~/.hermes/skills (or a subset) under git. Share skills across machines or team.

Manual creation

Create a folder + SKILL.md by hand, or use /learn to distill a workflow or past conversation.

Approval gate

Enable skills.write_approval so every new skill requires your confirmation.

Hub & community

Install skills from agentskills.io-compatible sources; audit before use.

Slash commands

Every skill becomes a /skill-name command automatically.

05 Use cases & managing skills on the filesystem

Use Cases

- **System administration**
- **Development workflows**
- **Scheduled automation**
- **Knowledge capture**
- **Multi-platform reach**
- **Team sidekick**

Package updates, service checks, log triage, zypper workflows turned into reusable skills.

Build, test, PR review, changelog generation — agent remembers your project conventions.

Built-in cron + natural language: daily reports, backups, weekly audits delivered to Telegram/email.

After solving a hard problem once, Hermes writes the procedure so the next run is near-instant.

Talk to the same agent from CLI, Telegram, Discord, or Slack while it works on your openSUSE box.

Specialist bots with their own skills/memory for ops, docs, or onboarding.

06 Hermes (pros & cons)

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Hermes Agent - Pros Cons

ADVANTAGES of Hermes

- Self-improving: skills compound over time
- Persistent memory & user model across sessions
- Skills are editable Markdown on your disk
- Full control of orchestration & tool policy
- Works with any LLM provider (no lock-in)
- Can drive real OS workflows on openSUSE
- Open source (MIT) — auditable & forkable
- Multi-platform gateway (chat + CLI)

TRADE-OFFS

- You operate the runtime (install, update, keys)
- Cloud LLM still incurs API cost & latency
- Skill quality depends on the runs that created them
- Less polished out-of-the-box chat UX
- Safety boundaries are your responsibility
- Smaller ecosystem than major cloud products
- Requires some Linux / admin familiarity
- Learning curve for skill curation

07 Conclusion

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Practical Blueprint on openSUSE

1	**Install**	curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
2	**Configure model**	hermes setup → point to OpenRouter / OpenAI / Anthropic / your endpoint
3	**First tasks**	Run real multi-step workflows; correct the agent when needed
4	**Watch skills appear**	Check ~/.hermes/skills/ — edit, improve, or git-track them
5	**Optional hardening**	Enable write approval, review tool permissions, limit sensitive paths
6	**Automate**	Use natural-language cron, messaging gateway, and specialist bots

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Key Takeaways

1. Hermes turns an openSUSE system into a self-improving AI orchestration layer.
2. The Learning Loop writes procedural knowledge as plain Markdown skills on disk.
3. You can use cloud LLMs while keeping skills, memory, and control local.
4. Skills are editable, shareable, and versionable — treat them like code.
5. Compared with pure Cloud AI like ChatGPT/Claude, you gain persistence and agency at the cost of operating the runtime.
6. Start small: install, connect a model, solve real tasks, then curate the skills that appear.

Thank You



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